Database Management Systems (DBMS) - 4 Module Syllabus

Module 1: Introduction to Database Systems

Objective: Learn the basics of databases and why they are used.

• Introduction to Data and Information
• File System vs Database System
• Database Concepts
• Advantages and Disadvantages of DBMS
• Components of DBMS
• Database Users and Roles
• Types of Databases (Relational, NoSQL, Distributed, Cloud)
• Database Architecture (1-Tier, 2-Tier, 3-Tier)
• Data Models (Hierarchical, Network, Relational, ER Model)

Module 2: Database Design and SQL

Objective: Learn how to design databases and work with SQL.

• ER Model and ER Diagrams
• Entities, Attributes, Relationships
• Keys (Primary, Foreign, Candidate, Composite)
• Mapping ER Model to Tables
• SQL Introduction
• DDL: CREATE, ALTER, DROP, TRUNCATE
• DML: INSERT, UPDATE, DELETE
• DQL: SELECT
• DCL: GRANT, REVOKE
• TCL: COMMIT, ROLLBACK, SAVEPOINT
• WHERE, ORDER BY, GROUP BY, HAVING, LIMIT
• Operators, Aggregate Functions
• Joins, Subqueries, Views, Indexes

Module 3: Relational Database Design

Objective: Design efficient and reliable databases.

• Functional Dependencies
• Database Anomalies
• Normalization (1NF, 2NF, 3NF, BCNF)
• Denormalization
• Referential Integrity
• Integrity Constraints
• Transactions and ACID Properties
• Concurrency Control
• Locking and Deadlocks
• Recovery Techniques
• Backup and Restore

Module 4: Advanced DBMS Concepts

Objective: Learn advanced database concepts.

• Query Processing and Optimization
• Storage Management and File Organization
• Indexing (B-Tree, Hash)
• Database Security
• Database Administration (DBA)
• Data Warehousing
• Data Mining
• NoSQL Databases
• Distributed Databases
• Cloud Databases
• Backup Strategies
• Case Study: Complete Database Design

Practical Lab Exercises

• Install MySQL or PostgreSQL
• Create databases and tables
• Perform CRUD operations
• Practice SQL queries
• Use aggregate functions and joins
• Write subqueries
• Create views and indexes
• Normalize a sample database
• Implement transactions
• Design ER diagrams for Library, Student, Hospital, and Online Shopping Systems

Learning Outcomes

• Understand DBMS fundamentals
• Design databases using ER models
• Write SQL queries confidently
• Normalize databases
• Manage transactions and concurrency
• Secure and administer databases
• Build complete relational database applications